

4.3 Relational Database Management System (RDBMS)

آر ڈی بی ایم ایس کئی ٹیبلز میں ڈیٹا رکھتا ہے اور کیز سے ان کے تعلقات برقرار رکھتا ہے۔

CS XI

Chapter 4: Data and Analysis

Lecture 05 of 15

4.3 Relational Database Management System (RDBMS)

LEARN IT

An **RDBMS** stores data in several tables and uses primary and foreign keys to keep relationships. Banks, universities and NADRA use RDBMS software. Extra strengths on top of a plain DBMS:

- Reduces redundancy by splitting data into related tables.
- Supports many users at once, with permissions.
- Is queried with **SQL** (Structured Query Language) — a standard language of SELECT, INSERT, UPDATE, DELETE.

WORKED EXAMPLE

Instead of one giant spreadsheet of students-and-departments, an RDBMS has Department(DeptID, DeptName) and Student(RollNo, Name, DeptID FK). CS is stored once.

SQL: SELECT Name FROM Student WHERE DeptID = 1;

اردو میں سمجھیں

آر ڈی بی ایم ایس کئی ٹیبلز میں ڈیٹا رکھتا ہے اور کیز سے ان کے تعلقات برقرار رکھتا ہے۔ ایس کیو ایل سے سوال پوچھے جاتے ہیں۔ اس سے تکرار کم اور کئی صارفین محفوظ طریقے سے کام کر سکتے ہیں۔

CHECK YOURSELF

1. What language do you use to query an RDBMS?
2. Why split Department out of the Student table?

Answers: 1. SQL. 2. To store the department name once (less redundancy) and update it in one place.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A DBMS is software that:

- a. Creates and manages databases
- b. Draws only pie charts
- c. Routes IP
- d. Sorts using bubble only

MCQ 2. Examples of DBMS:

- a. MS Access, MySQL, SQLite, Oracle
- b. Python only
- c. Figma only
- d. Chrome only

MCQ 3. DBMS handles:

- a. Storage, retrieval, security, multi-user access
- b. Only fonts
- c. Only OSI layer 1
- d. Only NAND

MCQ 4. Without a DBMS, programs would:

- a. Each invent their own file format
- b. Run faster always
- c. Need no keys
- d. Need no SQL

MCQ 5. Access is used in XI because:

- a. It is a GUI DBMS for tables/forms/queries/reports
- b. It is a programming language
- c. It is Agile
- d. It is analog

MCQ 6. SQLite in XII is a:

- a. Lightweight DBMS/library
- b. Chart
- c. Gate
- d. Sprint

Extra short questions

1. Define DBMS.
2. Four DBMS functions.
3. DBMS vs database.
4. Name two desktop and two server DBMS.

Extra long question

1. Compare managing 500 students in Excel vs a DBMS (Access). Use data integrity, search, multi-user and reports.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Database Management System: software to define, store, query, protect and share a database.
2. Create tables, insert/update/delete, query, enforce keys/security, backup.
3. Database is the data; DBMS is the software that manages it.
4. Desktop: Access, SQLite. Server: MySQL, Oracle, PostgreSQL.

Long

1. Excel: easy but duplicates, weak keys, clash when two people edit. DBMS: primary keys, queries, forms, reports, controlled access.

Next lecture: 4.4 Entity Relationship Model (ER-Model)